

Hand Calculation Guide

Sections 2.1 and 5.1 — Copy This on Paper

02417 Assignment 1

Section 2.1 — Matrix Form (write this on paper)

The linear trend model:

$$Y_t = \theta_1 + \theta_2 x_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2), \quad t = 1, \dots, N.$$

In matrix form for the first 3 time points:

$$\underbrace{\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{bmatrix}}_{\mathbf{X}} \underbrace{\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}}_{\boldsymbol{\theta}} + \underbrace{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}}_{\boldsymbol{\varepsilon}}$$

Inserting the actual values (3 decimal places):

$$\mathbf{y} = \begin{bmatrix} 2.930 \\ 2.934 \\ 2.941 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & 2018.000 \\ 1 & 2018.083 \\ 1 & 2018.167 \end{bmatrix}$$

So the full equation with numbers is:

$$\begin{bmatrix} 2.930 \\ 2.934 \\ 2.941 \end{bmatrix} = \begin{bmatrix} 1 & 2018.000 \\ 1 & 2018.083 \\ 1 & 2018.167 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}$$

That's it for 2.1. Write the general form, then plug in the numbers.

Section 5.1 — RLS Hand Calculation (write this on paper)

Step 0: Write the update equations

The RLS algorithm updates at each time step t :

$$\mathbf{R}_t = \mathbf{R}_{t-1} + \mathbf{x}_t \mathbf{x}_t^\top \quad (1)$$

$$\hat{\boldsymbol{\theta}}_t = \hat{\boldsymbol{\theta}}_{t-1} + \mathbf{R}_t^{-1} \mathbf{x}_t (Y_t - \mathbf{x}_t^\top \hat{\boldsymbol{\theta}}_{t-1}) \quad (2)$$

Initialize with:

$$\mathbf{R}_0 = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}, \quad \hat{\boldsymbol{\theta}}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Iteration $t = 1$

Given:

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2018 \end{bmatrix}, \quad y_1 = 2.930$$

Step 1 — Compute $\mathbf{x}_1 \mathbf{x}_1^\top$:

$$\mathbf{x}_1 \mathbf{x}_1^\top = \begin{bmatrix} 1 \\ 2018 \end{bmatrix} \begin{bmatrix} 1 & 2018 \end{bmatrix} = \begin{bmatrix} 1 \times 1 & 1 \times 2018 \\ 2018 \times 1 & 2018 \times 2018 \end{bmatrix} = \begin{bmatrix} 1 & 2018 \\ 2018 & 4\,072\,324 \end{bmatrix}$$

Step 2 — Compute $\mathbf{R}_1$:

$$\mathbf{R}_1 = \mathbf{R}_0 + \mathbf{x}_1 \mathbf{x}_1^\top = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} + \begin{bmatrix} 1 & 2018 \\ 2018 & 4\,072\,324 \end{bmatrix} = \begin{bmatrix} 1.10 & 2018.00 \\ 2018.00 & 4\,072\,324.10 \end{bmatrix}$$

Step 3 — Compute innovation:

$$y_1 - \mathbf{x}_1^\top \hat{\boldsymbol{\theta}}_0 = 2.930 - \begin{bmatrix} 1 & 2018 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 2.930 - 0 = 2.930$$

Step 4 — Compute $\hat{\boldsymbol{\theta}}_1$:

$$\hat{\boldsymbol{\theta}}_1 = \hat{\boldsymbol{\theta}}_0 + \mathbf{R}_1^{-1} \mathbf{x}_1 \cdot 2.930 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \mathbf{R}_1^{-1} \begin{bmatrix} 1 \\ 2018 \end{bmatrix} \cdot 2.930 = \begin{bmatrix} 0.0000001 \\ 0.001452 \end{bmatrix} \approx \begin{bmatrix} 0 \\ 0.00145 \end{bmatrix}$$

Iteration $t = 2$

Given:

$$\mathbf{x}_2 = \begin{bmatrix} 1 \\ 2018.083 \end{bmatrix}, \quad y_2 = 2.934$$

Step 1 — Compute $\mathbf{x}_2 \mathbf{x}_2^\top$:

$$\mathbf{x}_2 \mathbf{x}_2^\top = \begin{bmatrix} 1 \\ 2018.083 \end{bmatrix} \begin{bmatrix} 1 & 2018.083 \end{bmatrix} = \begin{bmatrix} 1 & 2018.083 \\ 2018.083 & 4\,072\,660.34 \end{bmatrix}$$

Step 2 — Compute $\mathbf{R}_2$:

$$\mathbf{R}_2 = \mathbf{R}_1 + \mathbf{x}_2 \mathbf{x}_2^\top = \begin{bmatrix} 1.10 & 2018.00 \\ 2018.00 & 4\,072\,324.10 \end{bmatrix} + \begin{bmatrix} 1 & 2018.083 \\ 2018.083 & 4\,072\,660.34 \end{bmatrix}$$

$$= \begin{bmatrix} 2.10 & 4036.08 \\ 4036.08 & 8\,144\,984.44 \end{bmatrix}$$

Step 3 — Compute innovation:

$$\mathbf{x}_2^\top \hat{\boldsymbol{\theta}}_1 = \begin{bmatrix} 1 & 2018.083 \end{bmatrix} \begin{bmatrix} 0.000001 \\ 0.001452 \end{bmatrix} = 1 \times 0.000001 + 2018.083 \times 0.001452 = 2.931$$

$$y_2 - \mathbf{x}_2^\top \hat{\boldsymbol{\theta}}_1 = 2.934 - 2.931 = 0.003$$

Step 4 — Compute $\hat{\boldsymbol{\theta}}_2$:

$$\hat{\boldsymbol{\theta}}_2 = \hat{\boldsymbol{\theta}}_1 + \mathbf{R}_2^{-1} \mathbf{x}_2 \cdot 0.003 = \begin{bmatrix} 0.000001 \\ 0.001452 \end{bmatrix} + \mathbf{R}_2^{-1} \begin{bmatrix} 1 \\ 2018.083 \end{bmatrix} \cdot 0.003 = \begin{bmatrix} 0.000000 \\ 0.001453 \end{bmatrix} \approx \begin{bmatrix} 0 \\ 0.00145 \end{bmatrix}$$

Comment: After just 2 observations the intercept $\hat{\theta}_1 \approx 0$ because with x -values around 2018, the slope $\hat{\theta}_2$ absorbs most of the signal. The estimates will stabilize as more data arrives. Note that we do not need to invert $\mathbf{R}_t$ by hand — the key point is to show the *structure* of the recursion.